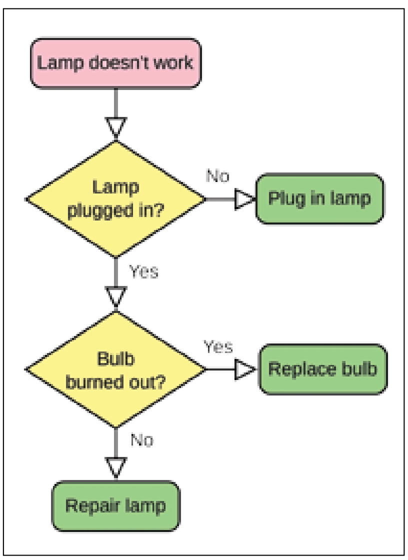
following machine learning course at http://dgit.in/MLStanfordCourse.

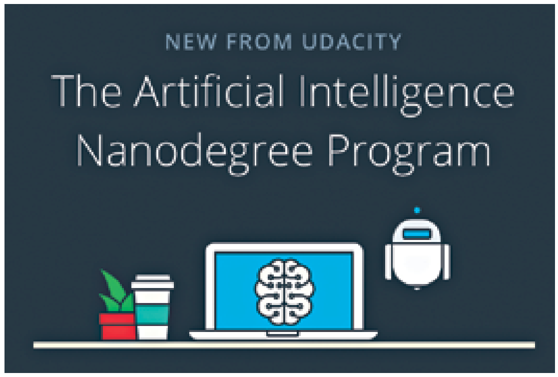
While there are more technical skills that you will need (ones that we are coming to) at this point, it is more important to talk about the non-technical skills that will be expected. Working in AI requires a constant urge to solve problems. Think of it like a riddle solving addiction – hobbies like game development or solving cross-word puzzles are generally a bonus.

Mathematics forms another important part of the required skillset. If you're not aware how machine learning works, it is understandable if you don't really understand the significance of mathematics in AI. However, in either case, a strong grasp in certain areas of mathematics is a must to solve actual problems with AI. A solid grasp of probability, statistics, linear algebra, mathematical optimisation, data sets and algorithms will be crucial in your understanding, and as a consequence, your construction of an AI system. In many other situations, treating an algorithm as a black box might not be a big deterrent but that isn't the case with AI.

If you're not working in research, working with AI will almost always involve the need for an industry expertise based on where it is being applied. For example, if you want to work at Waymo, expertise in automobiles is a bonus. In fact, a lot of AI based companies look at hiring experts in another field and then training them on the job in areas like machine learning and data mining. This is due to the lack of clear academic direction when it comes to treating artificial intelligence as a subject.



While this might be an oversimplification, dealing with logic in the form of flowcharts is something you will be doing a lot in AI work



Udacity also offers a lot of free courses in this area, along with the Nanodegree

As a result, the most reliable source from where you can learn more about AI related topics and develop skills is the internet. Platforms like Udacity and Coursera offer some really useful courses in this area. Udacity's Artificial Intelligence Nanodegree is developed in collaboration with IBM Watson, Amazon Alexa and Didi. It is a two-term program that first introduces the fundamentals of AI, then further allows you to focus on your area of interest. The areas include Natural Language Processing, computer vision, or speech recognition.

Working it out

Quite similar to the cloudiness surrounding the skills required for AI, there is a confusing mist that covers the kind of roles that you will get if you are looking for a job in artificial intelligence. To clear that up, let us start off by saying that the most common roles that you will see will be closely related to the skills that you've seen so far – machine learning and data science.

When it comes to the kind of responsibilities and the work that you'll have, this is where you will find quite a few familiar terms. Broadly, these are the areas you'll most likely be working in (along with some of the most prominent companies, both Indian and global, working in that area):

- **Natural Language Processing and generation**
  - Generation: Attivio, Automated Insights, Cambridge Semantics, Digital Reasoning, Lucidworks, Narrative Science, SAS, Yseop
  - Speech Recognition: NICE, Nuance Communications, OpenText, Verint Systems.
- **Computer vision** - Affectiva, Blippar, Blue River Technology, Captricity, uSens Solutions
- **Virtual agents/assistants:** Amazon, Apple, Artificial Solutions, Assist AI, Creative Virtual, Google, IBM, IPsoft, Microsoft, Satisfi
- **Machine learning:** Amazon, Fractal Analytics, Google, H2O.ai, Microsoft, SAS, Skytree
- **Deep learning:** Deep Instinct, Ersatz Labs, Fluid AI, MathWorks, Saffron Technology, Sentient Technologies.
- **Decision management:** Advanced Systems Concepts, Informatica, Maana, Pegasystems, UiPath
- **Robotic process automation:** Advanced Systems Concepts, Automation Anywhere, Blue Prism, UiPath, WorkFusion

While the global artificial intelligence industry has gone ahead by leaps and bounds, India is not too far behind. This pace is being achieved by startups who are doing some really innovative work in this area. "India is in the middle of an artificial intelligence revolution", explains Ishan Gupta, MD, Udacity India, "Along with the biggies, which are heavily investing into AI, more and more startups have come up that use or deal with artificial intelligence. Some of the companies in India that one should look out for are Google, Microsoft, LinkedIn, Adobe, Amazon, Ola, Flipkart and Myntra to name a few." Even though the major focus right